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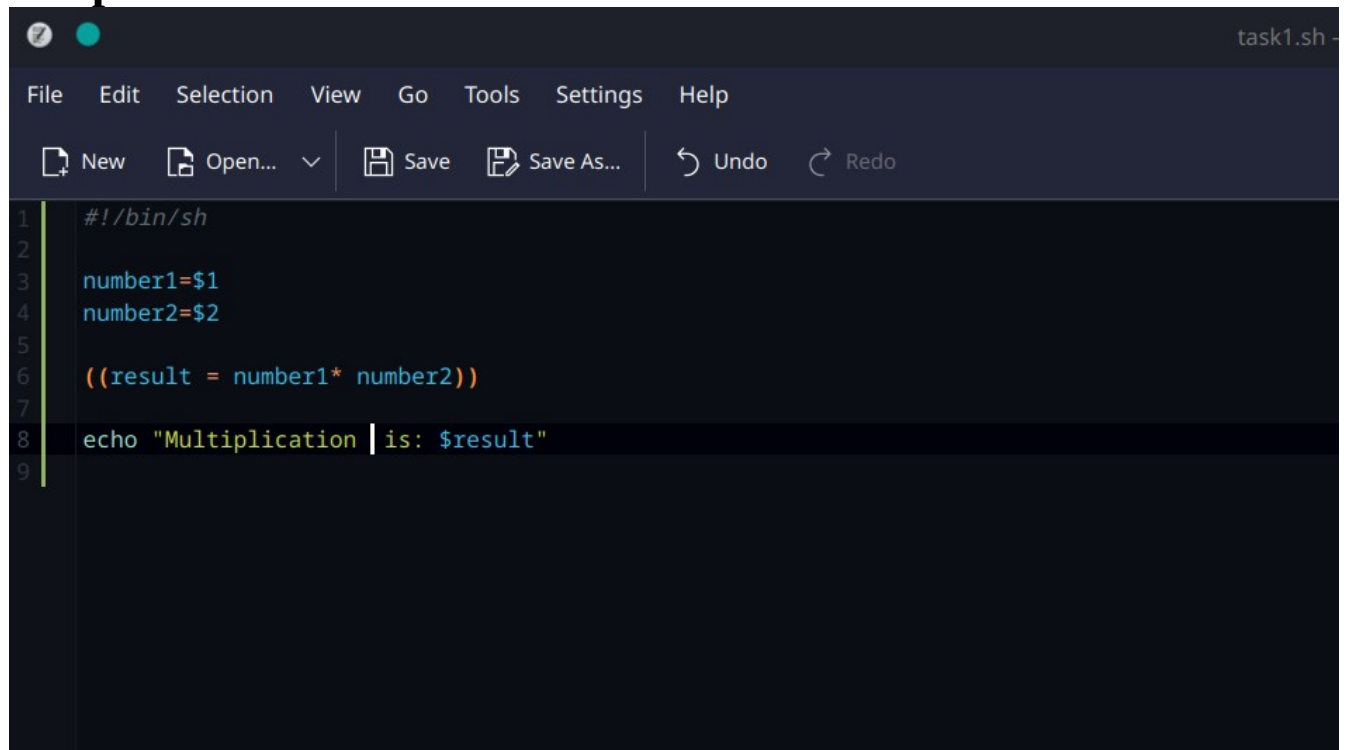
Section : BCS-4K

LAB TASK 3

1st 3 Questions

Task1:

Script:

A screenshot of a code editor window titled 'task1.sh'. The editor has a dark theme and a menu bar with 'File', 'Edit', 'Selection', 'View', 'Go', 'Tools', 'Settings', and 'Help'. Below the menu bar is a toolbar with icons for 'New', 'Open...', 'Save', 'Save As...', 'Undo', and 'Redo'. The code is written in a shell script format, starting with a shebang line '#!/bin/sh'. It then assigns values to 'number1' and 'number2' using '\$1' and '\$2'. A calculation is performed using '((result = number1 * number2))'. Finally, an 'echo' statement outputs the result with the text 'Multiplication is: \$result'. The code is displayed on a grid with line numbers from 1 to 9 on the left side.

```
1 #!/bin/sh
2
3 number1=$1
4 number2=$2
5
6 ((result = number1 * number2))
7
8 echo "Multiplication is: $result"
9
```

Result:

```
lab3task : bash — Konsole

File Edit View Bookmarks Plugins Settings Help

New Tab Split View

fkhokar@fedora:~/Desktop/lab3task$ chmod +x task1.sh
fkhokar@fedora:~/Desktop/lab3task$ ./task1.sh 2 3
Multiplication is: 6
fkhokar@fedora:~/Desktop/lab3task$
```

Task2: Script:

```
task2.sh —

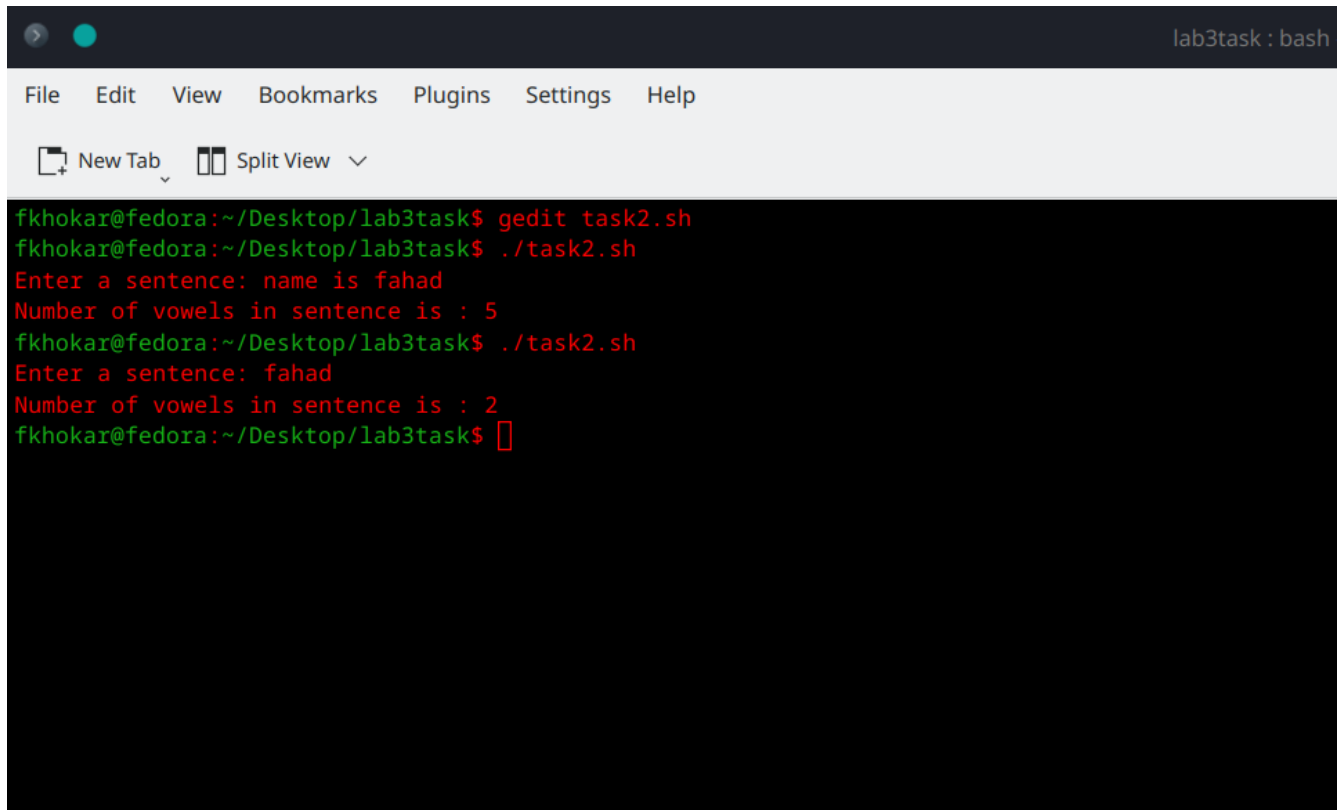
File Edit Selection View Go Tools Settings Help

New Open... Save Save As... Undo Redo

task1.sh task2.sh

1  #!/bin/sh
2
3  read -p "Enter a sentence: " sentence
4  counter=0
5
6  for ch in $(echo "$sentence" | sed 's/./& /g')
7  do
8      case $ch in
9          a|e|i|o|u|A|E|I|O|U)
10             ((counter++))
11             ;;
12      esac
13  done
14
15  echo "Number of vowels in sentence is : $counter"
16
17
```

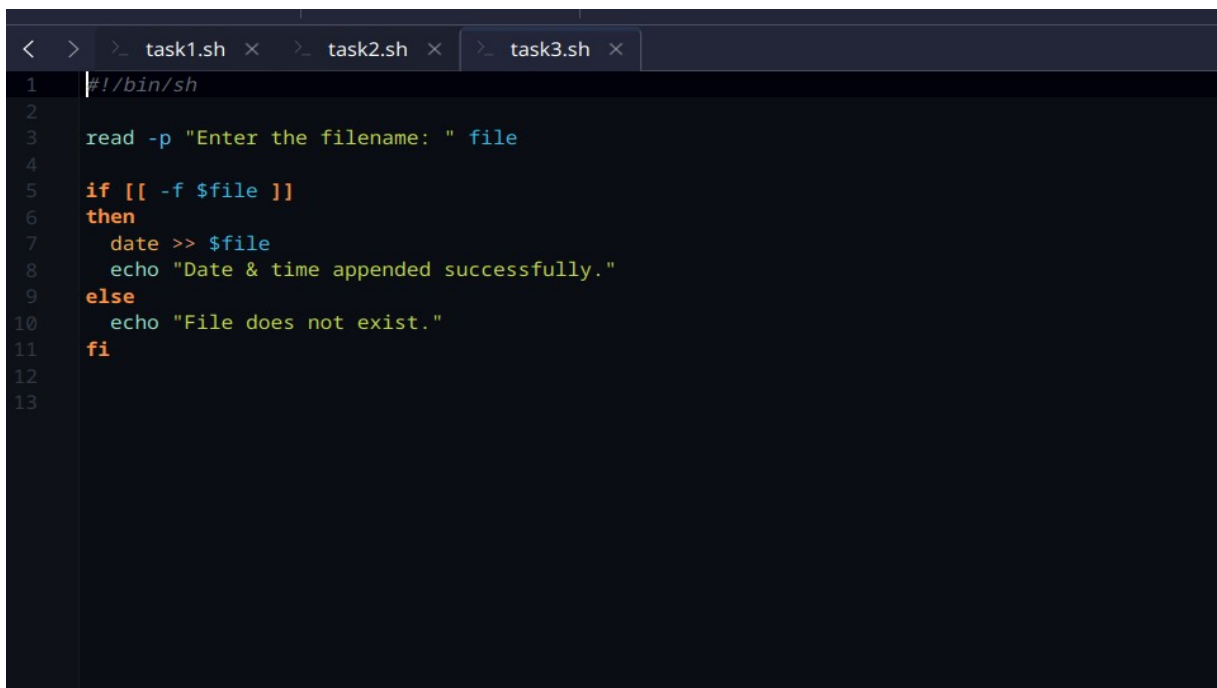
Result:



A screenshot of a terminal window titled "lab3task : bash". The window has a menu bar with "File", "Edit", "View", "Bookmarks", "Plugins", "Settings", and "Help". Below the menu bar are buttons for "New Tab" and "Split View". The terminal shows the following commands and output:

```
fkhokar@fedora:~/Desktop/lab3task$ gedit task2.sh
fkhokar@fedora:~/Desktop/lab3task$ ./task2.sh
Enter a sentence: name is fahad
Number of vowels in sentence is : 5
fkhokar@fedora:~/Desktop/lab3task$ ./task2.sh
Enter a sentence: fahad
Number of vowels in sentence is : 2
fkhokar@fedora:~/Desktop/lab3task$
```

Task3:



A screenshot of a code editor window with three tabs: "task1.sh", "task2.sh", and "task3.sh". The "task3.sh" tab is active, showing a shell script with the following content:

```
1  #!/bin/sh
2
3  read -p "Enter the filename: " file
4
5  if [[ -f $file ]]
6  then
7      date >> $file
8      echo "Date & time appended successfully."
9  else
10     echo "File does not exist."
11  fi
12
13
```

there is no file in this directory:

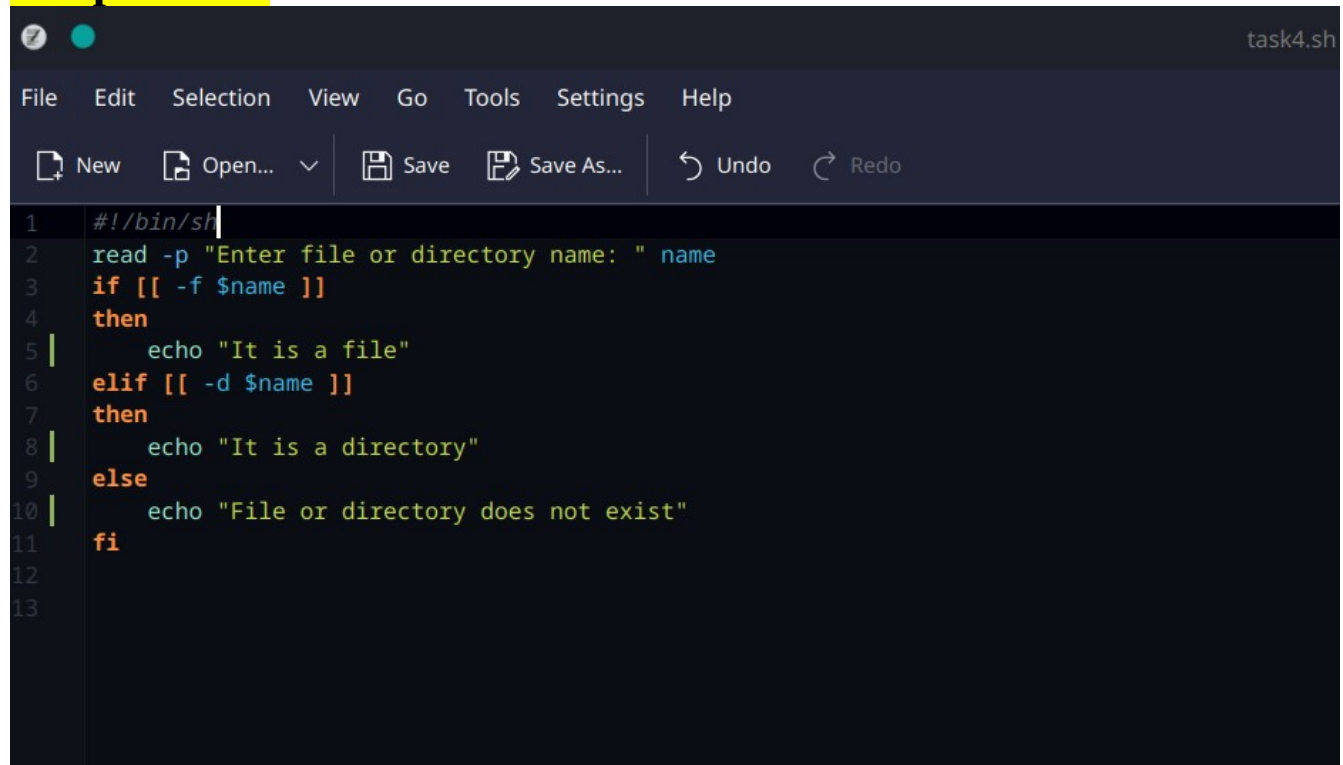
```
fkhoar@fedora:~/Desktop/lab3task$ gedit task3.sh
fkhoar@fedora:~/Desktop/lab3task$ chmod +x task3.sh
fkhoar@fedora:~/Desktop/lab3task$ t./task3.sh
bash: t./task3.sh: No such file or directory
fkhoar@fedora:~/Desktop/lab3task$ /task3.sh
bash: /task3.sh: No such file or directory
fkhoar@fedora:~/Desktop/lab3task$ ./task3.sh
Enter the filename: os.txt
File does not exist.
fkhoar@fedora:~/Desktop/lab3task$
```

Now here first I create then run the script

```
fkhoar@fedora:~/Desktop/lab3task$ touch os.txt
fkhoar@fedora:~/Desktop/lab3task$ echo "My os lab File" >> test.txt
fkhoar@fedora:~/Desktop/lab3task$ ./task3.sh
Enter the filename: os.txt
Date & time appended successfully.
fkhoar@fedora:~/Desktop/lab3task$ cat os.txt
Fri Feb 6 10:18:00 AM PKT 2026
fkhoar@fedora:~/Desktop/lab3task$
```

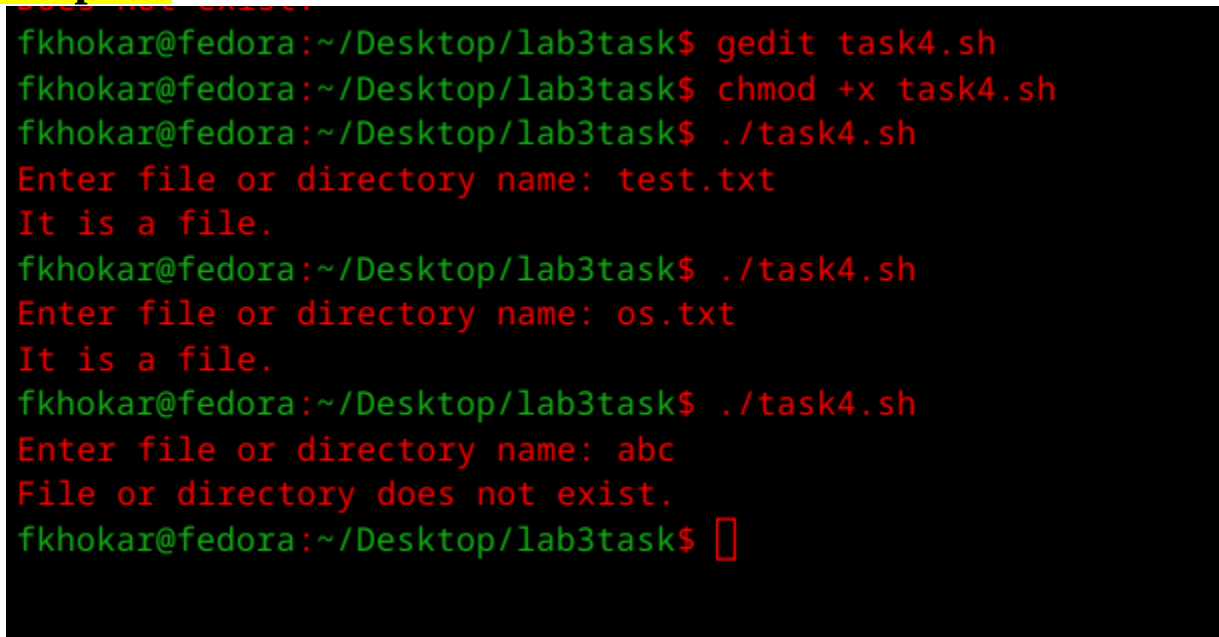
Task4:

Script Code:



```
task4.sh
File Edit Selection View Go Tools Settings Help
New Open... Save Save As... Undo Redo
1 #!/bin/sh
2 read -p "Enter file or directory name: " name
3 if [[ -f $name ]]
4 then
5     echo "It is a file"
6 elif [[ -d $name ]]
7 then
8     echo "It is a directory"
9 else
10    echo "File or directory does not exist"
11 fi
12
13
```

Output :



```
fkhoar@fedora:~/Desktop/lab3task$ gedit task4.sh
fkhoar@fedora:~/Desktop/lab3task$ chmod +x task4.sh
fkhoar@fedora:~/Desktop/lab3task$ ./task4.sh
Enter file or directory name: test.txt
It is a file.
fkhoar@fedora:~/Desktop/lab3task$ ./task4.sh
Enter file or directory name: os.txt
It is a file.
fkhoar@fedora:~/Desktop/lab3task$ ./task4.sh
Enter file or directory name: abc
File or directory does not exist.
fkhoar@fedora:~/Desktop/lab3task$
```

Task 5:

script code:

```
1  #!/bin/sh
2  sourcedirectory=$1
3  backupdirectory="backup_$(date +%Y-%m-%d)"
4
5  if [[ ! -d $sourcedirectory ]]
6  then
7      echo "Source directory does not exist"
8      exit
9  fi
10
11  mkdir $backupdirectory
12  cp -r $sourcedirectory $backupdirectory
13
14  files=$(find $sourcedirectory -type f | wc -l)
15  Directories=$(find $sourcedirectory -type d | wc -l)
16
17  echo "Backup completed successfully!"
18  echo "Backup Folder: $backupdirectory"
19  echo "Files backed up: $files"
20  echo "Directories backed up: $Directories"
21
22
```

Output:

```
fkhoar@fedora:~/Desktop/lab3task$ mkdir testdir
touch testdir/a.txt
fkhoar@fedora:~/Desktop/lab3task$ ls
os.txt task1.sh task2.sh task3.sh task4.sh task5.sh testdir test.txt
fkhoar@fedora:~/Desktop/lab3task$ chmod +x task5.sh
fkhoar@fedora:~/Desktop/lab3task$ ./task5.sh testdir
Backup completed successfully!
Backup Folder: backup_2026-02-07
Files backed up: 1
Directories backed up: 1
fkhoar@fedora:~/Desktop/lab3task$ ls
backup_2026-02-07 os.txt task1.sh task2.sh task3.sh task4.sh task5.sh testdir test.txt
fkhoar@fedora:~/Desktop/lab3task$
```